

Ashish Pal, Graduate Student

✉ ap81@rice.edu

🎓 Google Scholar

☎ +1 832-661-6807

Education

- 2019 – 2024 📖 **Ph.D. Civil and Environmental Engineering, Rice University** in Texas, USA
CGPA: 4.0/4.0
- 2013 – 2018 📖 **B.tech-M.Tech Integrated program. Civil Engineering, Indian Institute of Technology Kanpur** in UP, India.
Postgraduate CGPA: 10.0/10.0
Undergraduate CGPA: 9.2/10.0

Research Publications

Journal Articles

- 1 A. Pal, S. Bhowmick, and S. Nagarajaiah, "Physics-informed ai and ml-based sparse system identification algorithm for discovery of pde's representing nonlinear dynamic systems," *submitted to Proceeding of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 2024.
- 2 A. Pal and S. Nagarajaiah, "Damage detection and localization via full-field strain reconstruction using strain map basis and measurement," *submitted to Smart Structures and Systems*, 2024.
- 3 A. Pal and S. Nagarajaiah, "Data fusion based on short-term memory kalman filtering using intermittent-displacement and acceleration signal with a time-varying bias," *Mechanical Systems and Signal Processing*, vol. 216, p. 111 482, 2024.
- 4 A. Pal and S. Nagarajaiah, "Sparsity promoting algorithm for identification of nonlinear dynamic system based on unscented kalman filter using novel selective thresholding and penalty-based model selection," *Mechanical Systems and Signal Processing*, vol. 212, p. 111 301, 2024.
- 5 A. Pal, W. Meng, S. M. Bachilo, R. B. Weisman, and S. Nagarajaiah, "Subsurface damage detection via noncontact laser based surface level strain sensing smart skin with carbon nanotubes," *Engineering Structures*, vol. 284, p. 116 017, 2023.
- 6 A. Pal, W. Meng, and S. Nagarajaiah, "Deep learning-based subsurface damage localization using full-field surface strains," *Sensors*, vol. 23, no. 17, p. 7445, 2023.
- 7 W. Meng, A. Pal, S. M. Bachilo, R. B. Weisman, and S. Nagarajaiah, "Next-generation 2d optical strain mapping with strain-sensing smart skin compared to digital image correlation," *Scientific reports*, vol. 12, no. 1, p. 11 226, 2022.
- 8 A. Pal, W. Meng, and S. Nagarajaiah, "Hybrid method for full-field response estimation using sparse measurement data based on inverse analysis and static condensation," *Journal of Infrastructure Intelligence and Resilience*, vol. 1, no. 2, p. 100 017, 2022.
- 9 A. Pal and V. K. Gupta, "A note on spectral velocity approximation at shorter intermediate periods," *Soil Dynamics and Earthquake Engineering*, vol. 141, p. 106 422, 2021.
- 10 A. Pal and V. K. Gupta, "Peak factor-based modal combination rule of response-spectrum method for peak floor accelerations," *Journal of Structural Engineering*, vol. 147, no. 7, p. 04 021 095, 2021.

Conference Proceedings

- 1 A. Das, A. Pal, M. Sajeer, S. Nagarajaiah, and S. Mukhopadhyay, "Acceleration and strain data fusion technique for displacement estimation of dynamic system," in *submitted to Engineering Mechanics Institute Conference*, 2024.

- 2 A. Pal and S. Nagarajaiah, "Short-term memory kalman filter-based data fusion method using intermittent-displacement and acceleration with time-varying bias," in *submitted to Engineering Mechanics Institute Conference*, 2024.
- 3 W. Meng, A. Pal, S. Bachilo M., B. R. Weisman, and S. Nagarajaiah, "Accurate and refined strain-sensing smart skin (s₄) and its advantages over digital image correlation for 2d non-contact strain mapping," in *submitted to International Conference Proceedings Series: the 8th World Conference on Structural Control and Monitoring (8WCSCM)*, Springer, 2022.
- 4 A. Pal, W. Meng, and S. Nagarajaiah, "Deep learning for image segmentation and sub-surface damage detection based on full-field surface strains," in *submitted to Model Validation and Uncertainty Quantification, Volume 3: Proceedings of the 40th IMAC, A Conference and Exposition on Structural Dynamics 2022*, Springer, 2022.
- 5 A. Pal, W. Meng, and S. Nagarajaiah, "Sub-surface damage detection using deep learning algorithm based on full-field strain measurement," in *submitted to International Conference Proceedings Series: the 8th World Conference on Structural Control and Monitoring (8WCSCM)*, Springer, 2022.
- 6 A. Pal, A. Gaur, and S. Mukhopadhyay, "Modal identification and damage detection of railway bridges using time-varying modes identified from train induced vibrations," in *Model Validation and Uncertainty Quantification, Volume 3: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, Springer, 2020, pp. 405–411.

Research Experience

Reviewer (2019–present)

- Served as a reviewer for several reputed Journals, including Structural Control and Health Monitoring, Journal of Structural Engineering, Journal of Mechanical Systems and Signal Processing, Journal of Building Engineering, and The Third International Nonlinear Dynamics Conference.

Ph.D. (2019–2024)

- Explored the engineering application of the next-generation non-contact spectroscopic strain mapping using the S⁴ method as a non-destructive evaluation technique for subsurface damage detection.
- A damage detection and localization technique has been developed based on the measured strains and FE model of the structure to accurately localize damage at different levels of load/damage.
- A non-destructive evaluation tool is developed based on surface strain measurements and deep learning for image segmentation for damage detection and localization of invisible subsurface damage.
- A hybrid method is developed for dense response estimation utilizing the FE model of the system and coarse measurements from Digital Image Correlation, surpassing the sensor acquisition limits.
- A data-fusion technique utilizing the Kalman filter is developed to estimate real-time displacement measurements using intermittent displacement data and acceleration data with a time-varying bias.
- A sparsity promoting algorithm is developed for identification of nonlinear dynamic system based on Unscented Kalman Filter using novel selective thresholding and penalty-based model selection.
- Developed a B-spline-based framework using physics-guided machine learning that simultaneously fits the measured data and discovers the fundamental nonlinear ordinary/partial differential equations of 1D, 2D, and 3D solid and fluid mechanics problems.

Research Assitant (2018–2019)

- Performed modal analysis of a simulated milling operation using input-output and output-only time domain techniques to extract the dynamic properties of the machine during its operating conditions.
- Performed modal testing of face milling and end milling tools for a comparative study of extracted modal properties from time domain and frequency domain methods.

Research Experience (continued)

Masters Thesis (2017-2018)

- Proposed a modal combination rule which estimates the peak floor accelerations of a building without using input power spectrum density function in contrast to the existing methods.
- Proposed a simple expression to estimate the non-stationarity factor of floor response from non-stationarity factors and spectrum ordinates corresponding to modes of vibration of the building.

Teaching experience

- | | |
|-----------------------|---|
| Jan 2024 - Apr 2024 | ■ Mentor: Master's Student Guided a master's student working on seismic protection of buildings using base isolation. Scheduled weekly meetings to monitor the progress, suggest next goals and doubt-solving session when necessary. |
| Aug 2023 - Apr 2024 | ■ Mentor: Undergraduate Students Guided two undergraduate students working on a project to understand the dynamics of structures. Schedule weekly meetings to give assignments, doubt-solving, and ensure steady progress. |
| Jan 2023 - Apr 2024 | ■ Teaching Assistant: Strength of Materials Laboratory Meticulous laboratory management to ensure the involvement of students during laboratory experiments given limited apparatus. Also delivered lectures before each lab. |
| Aug 2021 - Dec 2023 | ■ Teaching Assistant: Steel and Concrete Laboratory Delivered lectures, planned, prepared, and ensured the smooth execution of lab experiments. Prepared assignments and graded them in a timely manner to give students the necessary feedback. |
| Aug 2017 - April 2018 | ■ Teaching Assistant: Finite Element Methods for Civil Engineering Applications, Structural Analysis, Structural Health Monitoring, Optimization Methods Prepared solutions to the assignments, conducted doubt-solving sessions and delivered tutorials. Managed the smooth conduction of quizzes and examinations, along with their proper evaluation. |
| September 2018 | ■ Teaching Assistant: Modal Analysis, Theory in Practice, TEQIP Workshop Demonstrated a laboratory experiment on roving hammer test of a freely suspended plate, along with a lecture on the same. Smoothly managed groups of visitors in the workshop to ensure their involvement and keep their interest. |

Relevant Courses

- | | |
|----------|---|
| Core | ■ Structural Dynamics, Non-Linear Finite Element Methods, System Identification with ML |
| Non-core | ■ Statistical machine learning, Computer Vision methods, Computational Photography |

Miscellaneous Experience

Awards and Achievements

- | | |
|-----------|---|
| 2015-2016 | ■ Academic Excellence Award at IITK for outstanding academic performance. |
| 2015 | ■ Overall best project at IITK for a working model of a Ceb Press among 55 other projects. |
| 2014 | ■ 1st Certificate of appreciation at IITK among 400 students for fabrication of rope maker. |

Scholarships

- | | |
|-----------|---|
| 2019-2023 | ■ SERB-Rice fellowship. Awarded by the Government of India and Rice matching fellowship. |
| 2013-2017 | ■ Merit Cum Means Scholarship. Awarded by the Government of India. |

Industrial Experience

Internee at AFCONS Infrastructure, India (April - May 2016)

- Designed a plate girder bridge considering several combinations of traffic loads. Created a model of a bridge on STAAD pro and conducted finite element analysis.
- Worked on the design of 13 bridges to be built in Kashmir, India.
- Provided the estimates of the type and size of crane required during construction.

Skills

Software Skills

- Language ■ C, C++, Python, PyTorch, LaTeX, VTeX
- Softwares ■ Maple, ABAQUS, Ansys, MATLAB, LS Dyna, SOLIDWORKS, NonLin, ArcGIS, Labview, CutPro, STAAD Pro, SAP2000

Experimental Skills

- Sensors ■ Worked with sensors and actuators such as triaxial accelerometers, hammers, dynamometers, laser vibrometers, spectrometers, and cameras.
- Testing ■ Dynamic testing of structure using methods like hammer test, and forced vibration test. Static/dynamic full-field strain sensing using advanced technologies such as Strain Sensing Smart Skin and Digital Image correlation on full-scale beams. Conducted several strength tests such as tensile, compression, and torsion tests.
- Machines ■ Experience with several machines for various tasks such as drilling, milling, and turning. One year of experience working with Computer Numerical Control (CNC) machining.

Extra Curricular

- Planned, and organized three presentation events in Techkriti 2016, leading a team of 10 members.
- Played Guitar at many Music Festivals and our band won 2nd prize in an inter-school competition.
- Won trophies in intra-hall cricket leagues for best bowling figures and highest runs in an inning.
- Was among top 30 players shortlisted for hockey summer training at IIT Kanpur in 2014.
- Won silver medal in 1500 m race during athletics meet at school.

References

- **Prof. Satish Nagarajaiah**
Professor in the Department of Civil and Environmental Engineering, and Mechanical Engineering, Rice University, Houston, Texas, USA, 77005.
E-mail: satish.nagarajaiah@rice.edu
- **Prof. Jamie Padgett**
Stanley C. Moore Professor of Engineering, and Professor in the Department of Civil and Environmental Engineering, Rice University, Houston, Texas, USA, 77005.
E-mail: jamie.padgett@rice.edu
- **Prof. R. Bruce Weisman**
Professor in the Department of Chemistry, Materials Science, and NanoEngineering, Rice University, Houston, Texas, USA, 77005.
E-mail: weisman@rice.edu